

MD ZAKARIA MASUD

Senior Software Engineer

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[GitHub](#) | [Linkedin](#) | [Portfolio](#)



SUMMARY

Senior Software Engineer with 5+ years of experience designing and building enterprise web applications, REST APIs, and cloud-native microservices on the .NET stack. Skilled in Clean Architecture, CQRS, and event-driven systems across Azure and AWS, with production experience spanning IoT platforms, ERP systems, and government infrastructure monitoring. Currently extending this background into Agentic AI and LLM-integrated applications using Microsoft Agent Framework, Azure AI Foundry, and MCP.

Areas of interest: distributed systems, cloud-native applications, software architecture, system design, agentic AI.

SKILLS

Languages: C#, TypeScript/JavaScript, SQL, Python, Java, C, PHP

Backend: ASP.NET Core (MVC, Web API, Razor Pages, EF Core, Dapper, SignalR), ASP.NET (MVC, Web API, Web Forms, EF), Quartz.NET, OutSystems, Laravel

Frontend: Angular, Angular Material, PrimeNG, React, HTML/CSS, Bootstrap, jQuery, Figma

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, SQLite

Architecture & Patterns: Clean Architecture, CQRS with MediatR, N-Layer Architecture, Repository & Unit of Work, Microservices, Event-Driven Design

Cloud & DevOps: Azure App Service, AWS (EKS Fargate, Lambda, S3, SQS, EventBridge), Docker, Kubernetes, RabbitMQ, EMQX MQTT, .NET Aspire, Git/GitHub/GitLab/Azure DevOps/TFS

AI & LLM Tooling: Prompt Engineering, RAG, Microsoft Agent Framework, Microsoft.Extensions.AI, Azure AI Foundry, MCP, Vector Databases, Agentic AI

EXPERIENCE

Kaz Software | Dhaka, Bangladesh

Senior Software Engineer
Software Engineer

Oct 2025 – Present
Oct 2024 – Sep 2025

- Own delivery of RESTful APIs in ASP.NET Core following Clean Architecture and CQRS, working across the full lifecycle from design through production deployment for local and international clients.
- Build and maintain enterprise web applications using ASP.NET Core, Angular, React, PostgreSQL, SQL Server, and MongoDB.
- Design serverless and background-processing solutions with Azure WebJobs, AWS Lambda, Amazon SQS, and EventBridge to support scalable backend services.
- Architect event-driven microservices using RabbitMQ for asynchronous communication and real-time notifications.
- Develop responsive front-end features with Angular, PrimeNG, and React.
- Deploy and manage workloads on Azure App Service and AWS (S3, serverless infrastructure), using Docker for containerization.
- Partner with cross-functional teams on requirements analysis, system design, code review, UAT, and production releases.
- Evaluate and integrate emerging AI tooling — Microsoft.Extensions.AI, Azure AI Foundry, Microsoft Agent Framework, and MCP — into existing software solutions.

S3 Innovate Pte. Ltd. | Midview City, Singapore (Remote)

Software Engineer L-II
Junior Software Engineer

Apr 2023 – Sep 2024
May 2021 – Mar 2023

- Developed REST APIs in ASP.NET Core Web API following Clean Architecture with CQRS and the Mediator pattern.
- Built and maintained .NET Worker Services powering backend microservices for an IoT device-monitoring platform.
- Developed ETL pipelines that processed IoT sensor data from an MQTT broker against configurable rule sets to generate alarms, events, and monitoring dashboards.
- Maintained SQL Server and PostgreSQL databases using Dapper and EF Core.
- Delivered front-end features with Angular, Angular Material, and Angular Flex-Layout, working from Figma designs.
- Automated scheduled jobs with Quartz.NET; containerized services with Docker and Kubernetes, deploying to AWS EKS Fargate.
- Researched advanced C# features, Kubernetes/Rancher, and API tooling to inform technical decisions; contributed to requirements analysis, UAT, and documentation.

- Authored SRS documentation and designed the PostgreSQL database for the PIM and VMS modules of an ERP system.
- Developed Web APIs in ASP.NET Core following Clean Architecture; tested APIs and managed version control via SourceTree.

- Designed the database schema in Microsoft SQL Server for the CMP Officer's Mess project and built the back end using ASP.NET MVC.
- Presented project deliverables directly to the client.

PROJECTS

CFEMS (*Official — S3 Innovate*): IoT equipment-monitoring platform for a Singapore government commuter-facility program. Built with .NET 6 (Web API, Worker Services, SignalR, EF Core), EMQX MQTT, Angular 12, Clean Architecture/CQRS with MediatR, and Kubernetes.

Reactivities (*Personal*): Full-stack social/activity-feed app built with ASP.NET Core Web API, Clean Architecture/CQRS/MediatR, Identity, SignalR, and a React + TypeScript + MobX front end. [Source](#)

BulkyBook (*Personal*): Online bookstore built with ASP.NET Core MVC, EF Core, Identity, and the Repository pattern, integrated with Stripe and Braintree payments. [Source](#) · [Live demo](#)

CERTIFICATIONS & TRAINING

- .NET Aspire and GenAI: Developing Distributed Architectures — *Udemy*
- ASP.NET Core MVC with Angular and EF Core — *PencilBox, Dhaka, Bangladesh, 2021*
- Complete Guide to ASP.NET Core MVC (v3.1) — *Udemy*
- Web Application Development – .NET — *BASIS Institute of Technology and Management (BITM), Chittagong, 2019*

[See more](#)

EDUCATION

BSc in Computer Science and Engineering

East Delta University, Chittagong, Bangladesh

2014 – 2019

Thesis: “DDoS Attack Detection with Protocol Type Using Machine Learning” — proposed a hybrid ensemble model combining five classifiers, achieving over 96.85% detection accuracy across three protocol types. Tools: Python, scikit-learn, Pandas, NumPy, Matplotlib.